

Proposal

The problem selected for this project is short-term solar power forecasting using deep learning. Specifically, the goal is to predict solar panel power output up to one hour into the future based on historical measurements and environmental conditions. This problem was chosen because accurate solar forecasting is critical for renewable energy integration, grid stability, and efficient energy storage management.

The dataset used for this project is a self-collected solar dataset recorded from January 28 to the present. The data includes measurements from solar panels and associated environmental variables such as irradiance, temperature, wind, and humidity. Because the data is continuously logged, it provides a time-series dataset with sufficient resolution and duration to train a deep learning model. While the dataset is smaller than large public datasets, its high temporal resolution and real-world consistency make it suitable for training sequence-based models like LSTMs.

The deep learning model selected for this project is a Long Short-Term Memory (LSTM) network. LSTMs are well-suited for time-series forecasting because they can capture temporal dependencies and long-term patterns in sequential data. The model will take a sequence of past observations (e.g., the previous several hours of data) as input and output predictions for future solar power over the next hour. The architecture will be a customized LSTM, including multiple layers and dropout for regularization, along with a fully connected output layer to produce multi-step forecasts.

The implementation framework used for this project will be PyTorch. PyTorch was chosen due to its flexibility, ease of debugging, and strong support for custom model architectures. It also integrates well with Python-based data processing tools such as NumPy and pandas, which are already being used for preprocessing and handling the collected dataset.

To support the development of this project, reference materials will include course lecture notes, PyTorch documentation, and research papers related to time-series forecasting and solar energy prediction. In particular, literature on LSTM-based forecasting and renewable energy prediction will be used to guide model design, training strategies, and evaluation methods.

The performance of the model will be evaluated using standard regression metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics will quantify how closely the predicted solar output matches the actual measured values. Additionally, the model's performance will be compared against a baseline method, persistence forecasting (assuming future values are equal to current values), to ensure that the deep learning model provides a meaningful improvement.

The rough schedule for completing the project is as follows:

- **Week 1:** Finalize dataset preprocessing and baseline model implementation
- **Week 2:** Design and implement the LSTM model
- **Week 3:** Train and tune the model (hyperparameters, architecture adjustments)
- **Week 4:** Evaluate performance and compare with baseline
- **Week 5:** Generate plots, analyze results, and prepare presentation materials
- **Week 6:** Write final report and finalize documentation